

How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h>
int
main()
{
    FILE *fp;    fp =
    fopen("sample.txt", "w");    if
    (fp == NULL)
    {        printf("Cannot open
    file\n");        return 1;
    }    fprintf(fp, "Hello, this is a test
    file.\n");    fclose(fp);    printf("File
    created successfully.\n");    return 0;
}
```

Compile `gcc permission.c -o permission`

Run

`./permission`

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read Others

have: Read

This is permission 644.

Give Read and Write permission to everyone `chmod`

```
666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt chmod  
o+rw sample.txt chmod a+rw  
sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read →
644
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
- 11) Owner: read + write; Group: read + write; Others:
nothing → 650
- 12) Everyone: read only → 444

```

ksriram@LAPTOP-COID71L1: ~$ ls
OS-lab hello.c
ksriram@LAPTOP-COID71L1:~$ cd OS-lab/2520090022_OSSP
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls
2520090022_Practical 2520090022_Skill ForgeOS
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ gedit permission.c

** (gedit:644): WARNING **: 04:59:11.998: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:644): WARNING **: 04:59:11.998: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ gcc permission.c -o permission
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ./permission
File created successfully.
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw-r--r-- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 666 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw-rw-rw- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod u+rw sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw-rw-rw- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 600 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw----- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 660 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw-rw---- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 644 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rw-r--r-- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 755 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rwxr-xr-x 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod u+rw sample.txt
chmod: cannot access 'chmod': No such file or directory
chmod: cannot access 'o+rw': No such file or directory
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod u+rw sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod o+rw sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rwxr-xrwx 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod u+x sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rwxr-xrwx 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod g+rw sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
-rwxrwxrwx 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ chmod 100 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ ls -l sample.txt
---x----- 1 ksriram ksriram 27 Aug 17 05:03 sample.txt
ksriram@LAPTOP-COID71L1:~/OS-lab/2520090022_OSSP$ |

```